

**K S Tharunpranav 251801359-2025-AIDS** ▾**K2****Started on** Thursday, 1 October 2026, 9:13 AM**State** Finished**Completed on** Thursday, 1 October 2026, 9:14 AM**Time taken** 49 secs**Marks** 1.00/1.00**Grade** **10.00** out of 10.00 (**100%**)

Question 1 | Correct | Mark 1.00 out of 1.00

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that $A[j] - A[i] = k$, $i \neq j$.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as $5 - 1 = 4$

So Return 1.

For example:

Input	Result
3 1 3 5 4	1

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2  #include <stdlib.h>
3
4  int main() {
5      int n, k;
6      if (scanf("%d", &n) != 1) return 0;
7      int *arr = (int *)malloc(n * sizeof(int));
8      if (arr == NULL) return 1;
9      for (int idx = 0; idx < n; idx++) {
10         if (scanf("%d", &arr[idx]) != 1) {
11             free(arr);
12             return 1;
13         }
14     }
15     if (scanf("%d", &k) != 1) {
16         free(arr);
17         return 1;
18     }
19     int i = 0;
20     int j = 1;
21     int found = 0;
22
23     while (i < n && j < n) {
24         if (i == j) {
25             j++;
26             continue;
27         }
28
29         int diff = arr[j] - arr[i];
30
31         if (diff == k) {
32             found = 1;
33             break;
34         } else if (diff < k) {
35             j++;
36         } else {
37             i++;
38         }
39     }

```

```
40     printf("%d\n", round);
41
42     free(arr);
43     return 0;
44 }
45
```

	Input	Expected	Got	
✓	3 1 3 5 4	1	1	✓
✓	10 1 4 6 8 12 14 15 20 21 25 1	1	1	✓
✓	10 1 2 3 5 11 14 16 24 28 29 0	0	0	✓
✓	10 0 2 3 7 13 14 15 20 24 25 10	1	1	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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